Night-Time Pressure Reduction and Burst Frequency

C. Fixture D. Fixture

Journal of Fixture Studies, 2019

FICTIONAL TEST MATERIAL: this paper does not exist; it was written for the citation-audit fixture. No author, utility, venue, or measurement described here is real.

Abstract

Water mains burst more often when they run at high pressure, and network pressure peaks at night, when demand is lowest. We report a two-year trial on a fictional utility in which pressure-reducing valves lowered night-time pressure by 20% in 12 districts. Burst frequency in those districts fell by 28% relative to the two years before the trial. Because every burst releases water until it is isolated, the fall in bursts also reduced the water lost to bursts. We describe the valve settings, the burst records, and the comparison with untreated districts.

1 Introduction

Each burst on a distribution main costs a repair, an interruption of supply, and the water that escapes before the main is isolated. Bursts are therefore a direct source of lost water. Pressure is one cause of bursts that an operator can control. At night, consumption drops and the pressure in a district rises towards the static head of its supply. Pressure-reducing valves can cut that night-time excess while leaving daytime service unchanged.

This paper asks a single question: how much does a sustained cut in night-time pressure reduce burst frequency? We answer it with a two-year trial on the fictional Fixture Plains network.

2 Method

2.1 Districts and valve settings

Twelve hydraulically isolated districts took part. Each is fed through a single pressure-reducing valve. Before the trial, every valve held a fixed outlet pressure. During the trial, the valves switched to a time-based profile that lowered the

outlet pressure by 20% between midnight and five in the morning, and restored the original setting for the rest of the day. Daytime pressure was not changed.

2.2 Burst records

A burst is a main break that required an unplanned repair. We counted bursts from the utility's repair records for the two years before the trial and for the two trial years, and normalised the counts by main length. Twelve districts with similar pipe age and material, whose valves were not changed, served as a comparison group.

3 Results

3.1 Burst frequency

Cutting night-time pressure by 20% reduced burst frequency by 28% across the 12 trial districts over the two-year trial, relative to the two years before it. The comparison districts showed no significant change over the same period, so the reduction is not explained by a mild winter or by a change in repair practice. The fall was present in both trial years and in 11 of the 12 districts.

3.2 Water lost to bursts

We estimated the water lost to each burst from the repair record and the recorded time until the main was isolated. Because the trial districts had fewer bursts, the estimated volume of water lost to bursts fell by 24% over the trial. The fall is slightly smaller than the fall in burst frequency, because the bursts that remained were, on average, on larger mains.

3.3 Service levels

No customer complaint of low pressure was traced to the night-time profile, and fire-flow tests at the critical point of each district remained within the required limits.

4 Limitations

The trial was designed to measure one effect, and its conclusions are limited to that effect.

This study does not examine leak localization: we did not attempt to find or pinpoint leaks, and the trial says nothing about how accurately leaks can be located. This study does not examine metering: no customer or district meters were installed or read for the trial, and the loss estimates rest on repair records rather than on metered flows. The trial ran for two years in twelve districts;

a longer record would be needed to separate the pressure effect from the slow ageing of the mains.